An Intelligent Framework for Legal Document Comparison and Verification

Dr. S. Amutha[1]
Professor , Department of CSE
Vel Tech Rangarajan
Dr.Sagunthala R&D Institute of
Science and Technology,
Chennai, India
dramuthas@veltech.edu.in

Sure Manusha[2]
Scholar , Department of CSE(AIDS)
Vel Tech Rangarajan
Dr.Sagunthala R&D Institute of
Science and Technology,
Chennai, India
vtu24167@veltech.edu.in

Pandaraboina Abhilash[3]
Scholar , Department of CSE(AIDS)
Vel Tech Rangarajan
Dr.Sagunthala R&D Institute of
Science and Technology,
Chennai, India
vtu22331@veltech.edu.in

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***Abstract*—Legal documents consisting of contracts, agreements, or certifications are considered the basis for many decision-making processes, rendering the integrity of the documents to be of primary importance. Due to the rise in digitalization in the field of law with its consequent growth in the number of digital documents, manual verification of the documents was considered to be a tedious, difficult, or inefficient process prone to errors due to the involvement of the manual factor. Especially detecting changes to the texts within the documents or determining the legitimacy of the signature associated with the documents is considered to be a tedious job by itself for the concerned professionals dealing with documents using the manual method. This article proposes an intelligent system for matching the nature of the documents with the aim of assisting the verification process by using the facility of artificial intelligence for document verification by including the aspects like natural language processing for the verification of the documents with the help of the signature associated with the document.The experimental evaluations performed using different sets of legal documents verify the superiority of the proposed approach in terms of high precision for finding similar contents as well as signature irregularities. The results confirm the potential of the system in developing an efficient solution for improving the efficiency, reliability, and robustness of the verification process for legal documents.**

***Index Terms*—Legal document verification, document similarity analysis, natural language processing, signature authentication, semantic text analysis, machine learning, artificial intelligence, legal informatics.**

I. Introduction

Legal documents include contracts, agreements, certificates, and affidavits; these are, in fact, the very backbone of any le- gal, financial, and administrative system. Ensuring authenticity and integrity of such documents is pivotal to avoiding fraud, unauthorized changes, and possible related litigations. The fast digitization of legal records and the ever-growing move toward electronic document exchange have turned traditional manual verification processes inefficient, very time-consuming, and prone to human error. Therefore, automated and intelligent methods that are able to verify legal documents with high accuracy and speed have become increasingly in demand.

Advances in AI have allowed the use of natural language processing and computer vision methodologies on document analysis. The semantic text similarity approach enables one to contrast the contents of documents beyond just superficial matches of keywords, which would allow for the detection of paraphrased or structurally modified text.Transformer-based language models achieve better performance with respect to contextual understanding of legally complex language. Human handwritten signature verification has been studied extensively using both machine learning and deep learning approaches, and convolutional neural networks achieve remarkable success in extracting discriminative visual features from signatures. Nevertheless, despite such advancements in automated document analysis, automated verification of legal documents still proves to be an daunting task, particularly since legal documents are highly structured and domain-specific in nature, where clauses can be very complex in both meaning and lengths. Furthermore, signature verification has proved to be an additional complex process with issues of intra-writer variation, inter-writer similarity, as well as skilled forgery. Typically, most of the existing systems address issues of semantic text analysis and signature verification individually or in isolation from each other, producing incomplete or unre- liable results of document verification in practical applications. The need for this paper is based on the shortcomings of the existing system by proposing an AI-based framework for semantic legal text similarity and handwritten signature veri- fication. The proposed framework can perform semantic text similarity for contractual clauses and authenticity for hand- written signatures by utilizing a transformer and CNN-based models, respectively. The proposed framework can perform comprehensive verification by utilizing joint text and signature authenticity. Thus, this proposed framework can perform more
reliable and comprehensive verification for documents.

II. Related Work

Automated analysis of legal documents has advanced significantly with the advent of natural language processing techniques, which are increasingly being used for the analysis of legal documents. Existing techniques analyzed documents using lexical techniques based on TF-IDF and Cosine Sim- ilarity, which are more limited compared to the semantic analysis being performed in the latest techniques, which are more efficient and robust to paraphrased statement changes, including the analysis

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performed by transformer techniques like BERT, which are more useful for the analysis of the content of legal documents, without checking the authenticity of the documents being used for the analysis.

Handwritten signature verification is one of those biometric verification problems that has been widely explored. In past decades, signature verification was performed using statistical models after extracting specific handwritten signature feature sets traditionally. More recently, signature verification was performed using powerful discriminative automatically learned image features via convolutional neural networks. These mod- els performed well across handwritten signature verification problems; moreover, they demonstrated greater immunity to both random and skilled forgeries. However, signature veri- fication methods generally operate independently from other document semantics. Despite some studies being conducted regarding hybrid or multi-modal document verification, the previously proposed methods lack a unified framework for verification of legal documents. For example, they do not verify the clauses at a semantic level. With this piece of work, we introduce a unified framework for document verification by considering both semantic document verification and signature verification. The novel contribution includes a unified frame- work for verifying documents by including both semantic and signature verification.

III. Problem Statement and Requirement analysis *A.*

Problem Statement

This exponential growth of electronic legal documents includes but is not limited to contracts, agreements, and certificates and compliance records. It has been a result of rapid digitization within workflows in the field of law. The authenticity and consistency of such documents need to be established, as even minor textual changes or forged signatures concerning such critical legal documents may lead to serious legal, monetary, and reputational losses. Traditional document verification for legality depends considerably on manual inspection by a legal expert, which in turn makes the entire process time-consuming, labour-intensive, and prone to human error, especially in large-scale repository document verification.

Most existing automated methods try to meet this challenge by either detecting textual similarity or performing signature verification as a standalone task. For text-based methods, keyword matching, and semantic similarity models usually do not detect paraphrased or contextually altered legal clauses reliably. On the other hand, approaches for signa- ture verification, if performed in isolation, cannot consider inconsistencies between document content and authenticity of signing. When it comes to text, things like keyword matching or semantic models often miss the mark—they struggle to catch paraphrased or subtly changed legal clauses. Flip to signature verification, and you've got another problem. If you only check the signature without looking at the document's content, you can miss inconsistencies or even clever forgeries that slip right through. Consequently, the central problem this research tries to solve arises from the fact that a unified, scalable, and intelligent framework for the verification of legal documents by assessing semantic content consistency and signature authenticity in combination is missing. Essentially, there is a need for an automated approach that would minimize reliance on manual verification while ensuring accurate, consistent, and depend- able results across diverse legal contexts.

B. Requirement Analysis

It needs to catch the true meaning behind the text, not just match keywords. In legal writing, rephrasing or just moving a clause can completely change what something means, so the system has to understand context and meaning at a deep level. On top of that, signature verification matters a lot. Signatures and seals often prove whether a document's real.

The system should spot forgeries whether someone tried to quickly scribble a fake, carefully copied the signature, or simulated it with sophisticated tools. It'll need deep learning to pick up on those tiny details, all while keeping false alarms—both false positives and negatives to a minimum. It alsoneeds to handle various document format types, such as image scans of PDFs and editable text files, among others.

To do this, it needs to clean up images and use optical character recognition (OCR), so it can turn scan-only documents into something machines can actually read, while keeping everything in place.

But here's the thing it shouldn't treat checking the text and analyzing signatures as totally separate steps. Instead, it should blend those results together and make one clear decision about whether the text is genuine. From a non-functional perspective, it needs to ensure that its design is both scalable and agile enough to analyze a potentially large number of documents with rapid response times, incorporating stringent security measures such as access controls on different user roles.

Besides that, it needs to ensure design flexibility to cope with constantly evolving forgery methods via its modular design architecture. Lastly, despite its high level of automation within its capabilities, its design needs to ensure that it is interactive to a degree to ensure both its accuracy and trustworthiness for forensics usage within the legal fraternity.

IV. Existing System and Limitations

A. Existing System

The existent system for the verification of different types of legal documents has been designed for the automation of some points for document verification. The great majority of existent verification methods are either text similarity verification or

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signature verification. The majority of text-based verification technologies use existent keyword matching technologies, different statistical measures for calculating similarities, as well as semantic embedding, so they can perform verification for two or more different types of legal documents and identify the common or duplicated text inside the documents. The existent verification technologies are developed based on natural lan- guage processing technologies, enabling the detection of legal document paraphrasing according to contextual relationships between clauses.

At the same time, signature verification systems have the aspiration to identify the authenticity of documents with the signature details obtained, which are hand-written or presented digitally. Existing methods normally identify the currently existing state-of-the-art signature verification systems, which might incorporate handcrafted representations of the characteristics present within the signature images, but the majority of the existing state-of-the-art are based on employing the CNN deep learning model to identify the different features present within the signature, which might be real or forged.

Several commercially available and academic studies suggest a set of methodologies or approaches toward partial automation in helping legal practitioners in the process of legal document analysis. However, these systems, in most cases, show similarity scores, classification, or verifications, which can help legal practitioners in the process of manual checking. This set of methodologies is partial in nature in helping legal practitioners in their activities of legal document analysis.

B. Limitations

Currently, different technologies as they exist for the verification of documents mostly revolve around ensuring one aspect or another of the overall document analysis process is somewhat more streamlined or even automated. Indeed, as one can observe, most of the different technologies developed thus far with regard to the overall concern of document analysis can mostly be categorized around one or another form of either textual or even signature verification analysis. Most forms of the overall notion of similarity revolve around different forms of either keyword or even semantic models that are geared towards allowing for some forms of common or even clauses within a given overall set of documents. Indeed, over the course of the past years or so, overall transformer models have been vastly advantageous, particularly with the bid towards allowing individuals a means of fully being able to understand even different forms of legal documents, thus being advanta- geous towards allowing for the delineation of any form of paraphrasing information. Besides that, different models with regard to signature verification, even CNN models themselves, exist and have indeed had their overall notion of fruition with a given form of established dataset within a manner of benchmarking.These limitations display a requirement for a unifying and scalable verification framework, which can effectively validate a given legal context both in relation to the semantic document content and the signatures.

V. PROPOSED METHODOLOGY

In this paper, an excellent methodology with excellent intelligent and integrated features has been instituted within the process of verifying legal documents depending on semantic textual similarity and autenticity of signatures. The major aim of this methodology is ensuring that with a minimum use of inspections, maximum accuracy exists within the process of verifying legal documents depending on certain features. Unlike other systems in use today, this method accepts both forms of verifications as a whole within the subject matter of considering semantic similarity and autenticity signatures separately from the rest.

The methodology will start with the acquisition of the documents. The documents may be in various forms because they may be provided in the form of scanned documents or even PDF and text files. Then there will be an element in the methodology concerning the preprocessing of documents with the objective of information extraction from the legal documents in the forms of text and graphics. The information contained in the document will be extracted depending on its semantic content by using the method of high-level information processing in the form of natural language processing. Similarly, the region for the signature extraction will be performed with the objective of validation using high-level image processing method.

Among what can be considered as integral issues of what is called the proposed methodology is that it is going to present an integration or fusion of what is determined from both the textual similarity component and signature verification component of that methodology so that it encourages that no falses will be present in verification, thus encouraging scalability in diverse legal spaces, particularly where they are used independently of each other.

VI. SYSTEM ARCHITECTURE

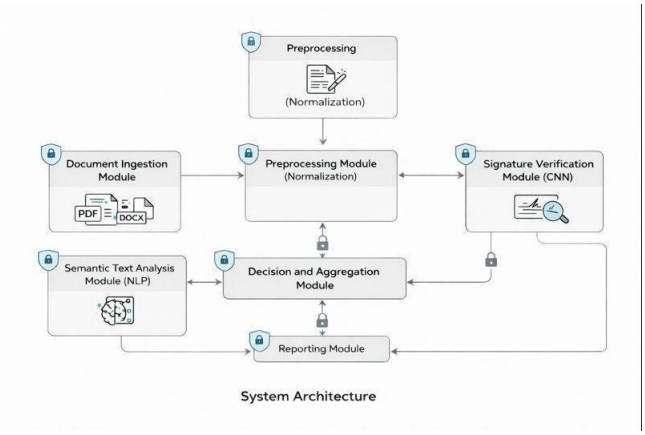


Fig. 1. Architecture of the proposed legal document similarity matching system with normalization-based preprocessing.

The proposal for legal document similarity matching is based on an architecture characterized by modularity, effi- ciency, and safety. In the proposed architecture, as depicted in Fig. 1, the legal

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documents are accepted in any form and are subjected to appropriate preprocessing such as OCR. Afterwards, the legal documents are analyzed by the semantic analysis and signature verification technologies in a parallel manner. In the proposed architecture, the legal documents are analyzed in terms of the textual similarity with the help of contextual embedding, while the legal documents are exam- ined in regard to the authenticity with the help of the analysis of the visual features. In the proposed architecture, the final output is presented with the help of the reporting layer.

VII. Implementation Modules

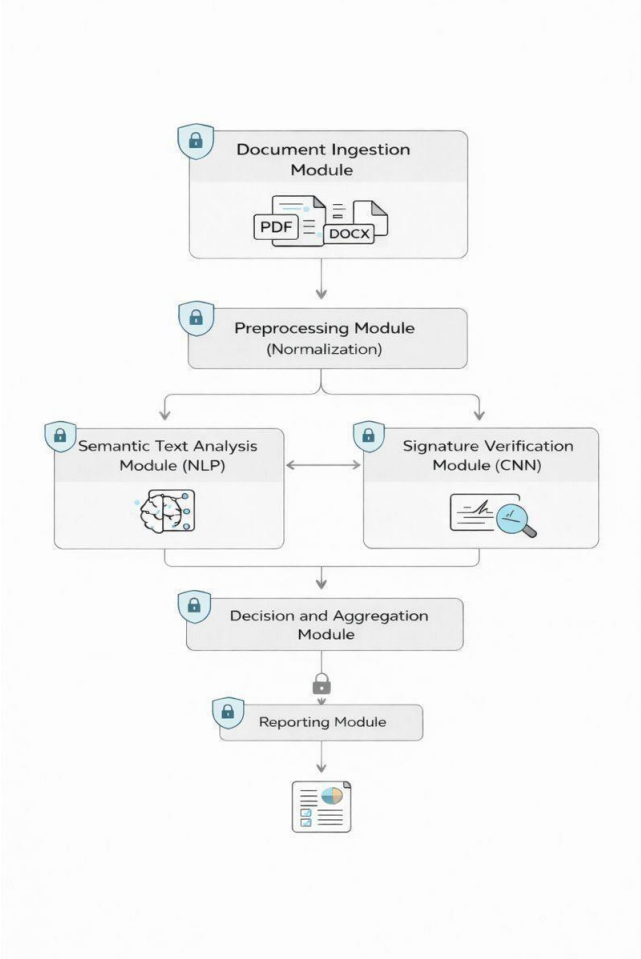


Fig. 2. Implementation modules of the proposed legal document similarity matching system.

The system under development would be implemented as modules that would be closely related to each other, as these would reflect the system architecture.The process of secure ingestion of legal documents in specific formats, like PDF and DOCX, is performed by the Document Ingestion Module. In the end, the input goes through additional processing stages. The text normalization process is performd by the Preprocessing Module by normalizing the text, thus making further stages effective. Normalization ensures standardization of text by eliminating any abnormalities in the text. Normalization of text is further passed through analysis by the Semantic Text Analysis Module in conjunction with NLP. This analysis is done by generating representations of text in the form of contexts in order to evaluate the similarity of text. Signature analysis is carried out by the Signature Verification Module in conjunction with CNN. This analysis is done in order to identify the accuracy of signatures. After successful identification, the results of these stages are concluded by the Decision and Aggregation Module. Thus, a report is compiled by Reporting Module in a compact form.

VIII. Result and Discussion

The experimental evaluation actually asserts the fact that the legal document similarity matching system approach out- ranks the traditional methods with the aid of this proposed framework. It has been analyzed taking into account the chal- lenges faced by the traditional approach while measuring legal document similarity through its reliance on both the similarity between the texts and authenticity of the signatures on the document, the approach to implement a semantic analysis on the similarity of texts coupled with the authenticity analysis on the signatures is useful for verifying the legal document itself.

IX. Implementation Modules

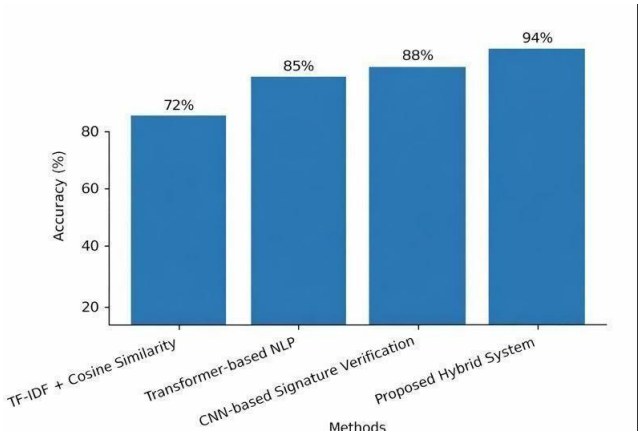


Fig. 3. Accuracy comparison of legal document verification methods.

If you look at the graph under “Implementation Modules,” it tells a pretty interesting story. It puts four techniques side by side, showing you what really matters: accuracy. The numbers aren’t just random: the accuracy jumps from 72% to 94%, and that’s no accident. It’s clear There is a strong connection here. The more complex and intelligent the models become, the better the outcomes. In other words, when you take the time to apply sophisticated methodologies, the legal document verification process.

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TF-IDF combined with Cosine Similarity gets you to about 72% accuracy, which isn't bad for something so straightforward. It mainly relies on word counting and calculates the similarity between the two documents on the basis of their counts. However, the reality is that legal documents cannot be defined based on word counts only; the technique fails to address the issue where comprehension of the meanings behind words becomes crucial. The meanings in legal texts can sometimes depend upon the manner in which they are framed, as well as the actual meanings behind the use of words.

The Signature Verification module based on CNN provides an entirely new perspective on the entire issue by focusing on the visual aspect of the signature and not merely its text or metadata. Indeed, legal documents are concerned with signatures, and there is nothing that could change a forged signature into a legitimate one no matter how skillfully it was done. Achieving accuracy of 88% definitely proves the effectiveness of implementing image analysis into authentication systems, since not only does it allow detecting forgery, but it protects from tampering too.

This leads to another milestone—the Proposed Hybrid System. Its success rate goes even higher at an impressive 94%. In this system, not only one technique operates independently; it blends several elements including textual analysis, contextual interpretation, and signature authentication into one efficient process. This is where things get exciting. Multiple levels are involved, such as verifying that the text makes sense, that it is suitable in terms of context and finally that the signature is legitimate. The effectiveness of the proposed system significantly reduces the chances of errors, which becomes very important when dealing with legal and other official documents. Take a look at the performance chart, and you will see why it stands out. There is absolutely no doubt that the hybrid model outperforms the previous ones by a significant margin, since combining different types of intelligence brings about validation that simply cannot be achieved with one type alone.

X. CONCLUSION AND FUTURE WORK

In this study, we developed a system to verify the authenticity of legal documents. We didn't just stop at one step; we put together several pieces — first, we clean up the documents to make their formats consistent, then we dive deeper and analyze their meaning, and finally, we verify the signatures. The results really highlight how much this integrated approach can help both in catching errors and in reducing the usual hassle people face with document verification.

To really put the Legal Document Similarity Matching System While going through its paces, various legal documents including contracts, agreements, certificates, among others, were used. However, this was not the only type of document used during the process, as the database was carefully designed to be reflective of actual cases. This meant that apart from using the genuine legal documents, their paraphrases, alterations, and even forgeries were also used. The forgeries in some instances went further and involved faking signatures in order to make them as realistic as possible. It had to be comprehensive because the system is supposed to perform flawlessly in any circumstances.

For the purposes of actually analyzing documents, however, we opted for using state-of-the-art Natural Language Processing tools, particularly models based on transformers. In all honesty, transformer models can do wonders when it comes to interpreting legal language nuances. Traditional approaches, where only word-by-word similarities would be searched, were completely out of the question; transformers can go much further than that, being capable of recognizing clause relations, detecting paraphrasing, identifying transitions between legal language forms, and spotting overlapping and contradictory clauses.

However, our efforts did not end there. In addition to analyzing text, the system we implemented also takes advantage of Convolutional Neural Networks. Consider CNNs as highly trained vision – they excel at pattern recognition and anomaly detection. In the context of this project, a CNN was specially developed to help identify signatures. Rather than simply looking for similarities in signatures, the neural network will identify patterns in the strokes used, their structure, and general composition.

The greatest advantage of our methodology is that everything is interconnected. The model itself does not solely analyze text but merges NLP for comparing texts with CNNs to verify signatures. Such an integration makes the entire process more reliable and complete. At the same time, while NLP focuses on identifying similarities and differences between texts, CNN concentrates on verifying signatures.

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